

**On Quantifying Energy Conversion Processes in Rotating Detonation
Combustors**

by

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APPENDIX C

Finite Volume Method Solver For Quasi-1D Flows

In this appendix, a one-dimensional finite volume method (1D-FVM) solver is outlined that includes the effects of area changes, friction losses, and heat release due to chemical reactions. The goal is simply to present the mathematics necessary to implement a working solver without focusing too much on the details. Most of the information presented here is taken from references [167, 168, 169, 65].

C.1 Conservation Laws

The general conservation equations for quasi-1D flow can be written in *conservative form* as follows

$$\frac{\partial}{\partial t} \begin{bmatrix} \rho A \\ \rho u A \\ \rho E A \end{bmatrix} + \frac{\partial}{\partial x} \begin{bmatrix} \rho u A \\ (\rho u^2 + p) A \\ u(\rho E + p) A \end{bmatrix} = \begin{bmatrix} 0 \\ p \frac{dA}{dx} - \frac{1}{2} \rho u^2 A \cdot \frac{4c_f}{D_h} \\ \dot{q} A - \frac{1}{2} \rho u^3 A \cdot \frac{4c_f}{D_h} \end{bmatrix} \quad (\text{C.1})$$

or more compactly as

$$\frac{\partial \bar{U}}{\partial t} + \frac{d\bar{F}(\bar{U})}{dx} = \bar{J} \quad (\text{C.2})$$

where $\bar{F}(\bar{U}) = Af(\bar{U})$, c_f is the skin friction coefficient, D_h is the hydraulic diameter, and $\dot{q}$ [W/m³] is the volumetric heat release which can be written as

$$\dot{q}A = -A \sum_k h_k \dot{\omega}_k \quad (\text{C.3})$$

where h_k is the mass specific enthalpy of species k given by

$$h_k(T) = h_f^\circ + \int_{T_{\text{ref}}}^T c_p dT \quad (\text{C.4})$$

and $\dot{\omega}_k$ is the source term for species k , which for steady flow is related to the mass fraction, Y_k , by

$$\frac{\partial}{\partial x}(\rho u Y_k A) = \dot{\omega}_k A \quad (\text{C.5})$$

Combining Eqs. (C.3) and (C.5) we obtain

$$\dot{q}A = - \sum_k h_k \left[Y_k \frac{d\dot{m}}{dx} + \dot{m} \frac{dY_k}{dx} \right] = -\dot{m} \sum_k h_k \frac{dY_k}{dx} \quad (\text{C.6})$$

Note that although Eq. (C.1) is written for unsteady flows, we are only interested in steady state solution and retaining the time varying terms allows the implementation of time marching numerical schemes.

We now express these equations in a form that is more convenient to solve. Define the variables $U_1 = \rho A$, $U_2 = \rho u A$ and $U_3 = \rho E A$ such that

$$\rho = \frac{U_1}{A} \quad (\text{C.7})$$

$$u = \frac{U_2}{U_1} \quad (\text{C.8})$$

$$p = \frac{\gamma - 1}{A} \left(U_3 - \frac{1}{2} \frac{U_2^2}{U_1} \right) \quad (\text{C.9})$$

$$T = \frac{\gamma - 1}{R} \left[\frac{U_3}{U_1} - \frac{1}{2} \left(\frac{U_2}{U_1} \right)^2 \right] \quad (\text{C.10})$$

and

$$\frac{\partial}{\partial t} \begin{bmatrix} U_1 \\ U_2 \\ U_3 \end{bmatrix} + \frac{\partial}{\partial x} \begin{bmatrix} U_2 \\ \frac{3 - \gamma}{2} \frac{U_2^2}{U_1} + (\gamma - 1) U_3 \\ \left[\gamma U_3 - \frac{1}{2} (\gamma - 1) \frac{U_2^2}{U_1} \right] \frac{U_2}{U_1} \end{bmatrix} = \begin{bmatrix} 0 \\ (\gamma - 1) \left(U_3 - \frac{1}{2} \frac{U_2^2}{U_1} \right) \frac{d}{dx} (\ln(A)) - \frac{1}{2} \frac{U_2^2}{U_1} f^* \\ \dot{q}A - \frac{1}{2} \frac{U_2^3}{U_1^2} f^* \end{bmatrix} \quad (\text{C.11})$$

where

$$f^* = \frac{4c_f}{D_h} \quad (\text{C.12})$$

The volumetric heat release term can be expressed similarly as

$$\dot{q}A = -U_2 \sum_k h_k \frac{dY_k}{dx} \quad (\text{C.13})$$

C.2 1D Finite Volume Method

The 1D finite volume method is a numerical technique for solving conservation laws. The domain is divided into control volumes, and the method evolves cell-averaged states over time by enforcing integral conservation within each cell. For a general partial differential equation of the form

$$\frac{\partial u}{\partial t} + \frac{\partial F(u)}{\partial x} = 0 \quad (\text{C.14})$$

The integral form is given by

$$\frac{d}{dt} \int_{x_L}^{x_R} u dx = F(u)|_{x_L} - F(u)|_{x_R} \quad (\text{C.15})$$

Applying Eq. (C.15) to cell j shown in Fig. C.1 yields

$$\frac{d}{dt} \int_{x_{j-1/2}}^{x_{j+1/2}} u dx = F(u)|_{x_{j-1/2}} - F(u)|_{x_{j+1/2}} \quad (\text{C.16})$$

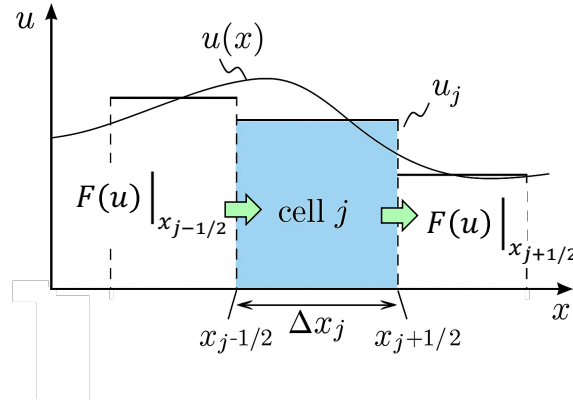


Figure C.1: Schematic of finite volume method problem.

We now define the cell-averaged state u_j as

$$u_j = \frac{1}{\Delta x_j} \int_{x_{j-1/2}}^{x_{j+1/2}} u dx \quad (\text{C.17})$$

and rewrite Eq. (C.16) as

$$\Delta x_j \frac{du_j}{dt} + F_{j+1/2} - F_{j-1/2} = 0 \quad (\text{C.18})$$

where $F_{j+1/2}$ is a shorthand notation for $F(u)|_{x_{j+1/2}}$. To implement a numerical method, we

must now approximate $F_{j+1/2}$ by a numerical flux $\hat{F}_{j+1/2}$ that we can evaluate at the cell interfaces $x_{j\pm 1/2}$ in terms of the cell-averaged states, u_j . Using a forward difference for the temporal derivatives we can now write for Eq. (C.2) the following:

$$\bar{U}_j^{n+1} = \bar{U}_j^n - \frac{\Delta t}{\Delta x} \left(\hat{F}_{j+1/2} - \hat{F}_{j-1/2} \right) + \Delta t \bar{J}_j \quad (\text{C.19})$$

C.3 Roe Flux

The Roe flux is used to compute the fluxes at cell interfaces by evaluating a linearized solution to the Riemann problem between neighboring cells. The Roe flux is constructed using the eigenvalues (λ_k) and right eigenvectors ($\tilde{r}_k$) of the Jacobian matrix of the flux function, evaluated at a Roe-averaged state. These eigenvalues represent wave speeds, and the eigenvectors define the wave structure, allowing the method to accurately resolve shocks, contact discontinuities, and expansion waves. The Roe flux is mathematically given by

$$\hat{F}_{j+1/2} = \frac{1}{2} A_{j+1/2} \left(\hat{f}_L + \hat{f}_R - \sum_k |\lambda_k| \tilde{\alpha}_k \tilde{r}_k \right) \quad (\text{C.20})$$

L and R indicate the left and right neighboring cells, and $\tilde{\alpha}_k$ are the wave strengths. Note that the entire flowfield needs to be initialized at the start in order to compute F_L and F_R . The Roe-averaged states ($\tilde{\bullet}$) are given by

$$r = \sqrt{\frac{\rho_R}{\rho_L}} \quad (\text{C.21})$$

$$\tilde{\rho} = \sqrt{\rho_L \rho_R} \quad (\text{C.22})$$

$$\tilde{u} = \frac{u_L + r \cdot u_R}{1 + r} \quad (\text{C.23})$$

$$\tilde{H} = \frac{H_L + r \cdot H_R}{1 + r} \quad (\text{C.24})$$

$$\tilde{a} = \sqrt{(\gamma - 1) \left(\tilde{H} - \frac{1}{2} \tilde{u}^2 \right)} \quad (\text{C.25})$$

where $H = E + p/\rho$. Note that if chemical reactions are modeled, then γ varies as a function of x , and therefore $\gamma_L \neq \gamma_R$. In this case, define a Roe-averaged squared speed of sound as

$$\tilde{a}^2 = \frac{a_L^2 + r \cdot a_R^2}{1 + r} \quad (\text{C.26})$$

and obtain $\tilde{a} = \sqrt{\tilde{a}^2}$. We now define the following differences

$$\delta\rho = \rho_R - \rho_L, \quad \delta u = u_R - u_L, \quad \delta p = p_R - p_L \quad (\text{C.27})$$

The Roe flux can now be computed using the following equations

$$\lambda = [\tilde{u} - \tilde{a}, \tilde{u}, \tilde{u} + \tilde{a}] \quad (\text{C.28})$$

$$\tilde{r}_1 = \begin{bmatrix} 1 \\ \tilde{u} - \tilde{a} \\ \tilde{H} - \tilde{u}\tilde{a} \end{bmatrix}, \quad \tilde{r}_2 = \begin{bmatrix} 1 \\ \tilde{u} \\ \frac{1}{2}\tilde{u}\tilde{u} \end{bmatrix}, \quad \tilde{r}_3 = \begin{bmatrix} 1 \\ \tilde{u} + \tilde{a} \\ \tilde{H} + \tilde{u}\tilde{a} \end{bmatrix} \quad (\text{C.29})$$

$$\tilde{\alpha}_1 = \frac{1}{2\tilde{a}^2}(\delta p - \tilde{\rho}\tilde{a}\delta u), \quad \tilde{\alpha}_2 = \delta\rho - \frac{\delta p}{\tilde{a}^2}, \quad \tilde{\alpha}_3 = \frac{1}{2\tilde{a}^2}(\delta p + \tilde{\rho}\tilde{a}\delta u) \quad (\text{C.30})$$

In some instances, an *entropy fix* is necessary to maintain an entropy satisfying solution and can be easily implemented as

$$|\lambda_k| = \begin{cases} \frac{\lambda_k^2 + \epsilon^2}{2\epsilon}, & |\lambda_k| < \epsilon, \\ |\lambda_k|, & \text{otherwise} \end{cases} \quad (\text{C.31})$$

where ϵ is a small number $\epsilon \approx 0.05\tilde{a}$. Regarding time marching, to satisfy the CFL condition, the time step can be chosen to be

$$\Delta t = \text{CFL} \cdot \min \left(\frac{\Delta x}{|u| + a} \right) \quad (\text{C.32})$$

C.3.1 Modeling Chemical Reactions

To include the effects of heat release due to chemical reactions, the spatial variation of species mass fractions (dY_k/dx) needs to be modeled. From Chapter 5 for H_2 -air, this can be achieved by specifying how the progress variable ξ varies spatially. One approach is to specify combustion efficiency at some location x_f and allow ξ to vary linearly from 0 at x_0

$$\xi = \eta_c \Phi \left(\frac{x - x_0}{x_f - x_0} \right) \quad (\text{C.33})$$

to specify combustion efficiency relative to the amount of fuel that can be burned for fuel rich conditions, Φ in Eq. (C.33) should be omitted. Species mole fractions are given by

$$X_{\text{H}_2\text{O}} = \frac{\xi}{\Phi + (1 - \xi)/2 + 1.88} \quad (\text{C.34})$$

$$X_{\text{H}_2} = \frac{\Phi - \xi}{\Phi + (1 - \xi)/2 + 1.88} \quad (\text{C.35})$$

$$X_{\text{O}_2} = \frac{(1 - \xi)/2}{\Phi + (1 - \xi)/2 + 1.88} \quad (\text{C.36})$$

$$X_{\text{N}_2} = \frac{1.88}{\Phi + (1 - \xi)/2 + 1.88} \quad (\text{C.37})$$

The mixture molecular weight can be determined using

$$\overline{M}_{\text{mix}} = \sum_k X_k \overline{M}_k \quad (\text{C.38})$$

and finally the mass fractions are given by

$$Y_k = X_k \cdot \frac{\overline{M}_k}{\overline{M}_{\text{mix}}} \quad (\text{C.39})$$

Other gas properties such as $\gamma(x)$, $R(x)$, and $h(x)$ can be obtained using NASA9 polynomial data and should be computed at each iteration / time step since they also depend on temperature.

C.3.2 Boundary Conditions

A summary of the number of variables that need to be specified at the inlet and outlet is provided in Table C.1. As an example, for a converging-diverging nozzle, with a subsonic inlet and supersonic outlet, we can choose to fix temperature and pressure at the inlet ($j = 1$). In this case, we first determine density using $\rho = p/RT$ and set $U_1^{j=1} = \rho A$. We now extrapolate $U_2^{j=1} = 2U_2^{j=2} - U_2^{j=3}$. This sets the inflow velocity $u = U_2^{j=1}/U_1^{j=1}$. To determine the value of $U_3^{j=1} = \rho u E$, we use $E = RT/(\gamma - 1) + u^2/2$.

C.4 Examples

Example plots of converged solutions for converging-diverging nozzles with known solutions obtained using Area-Mach number relations are shown in Figs. C.2 and C.3. Fig. C.4 shows the results for a simulation of a H₂-air rocket. Chemical reactions are modeled in combustor

Table C.1: Summary of flow boundary conditions.

Flow Regime	Inflow	Outflow
Subsonic	Fix two primitive variables and extrapolate the third.	Fix one primitive variable (usually pressure) and extrapolate the remaining two.
Supersonic	Fix all variables. The interior points have no influence.	Extrapolate all variables.

($-0.5 \leq x \leq 0$) by allowing ξ to vary linearly. The flow is then ideally expanded.

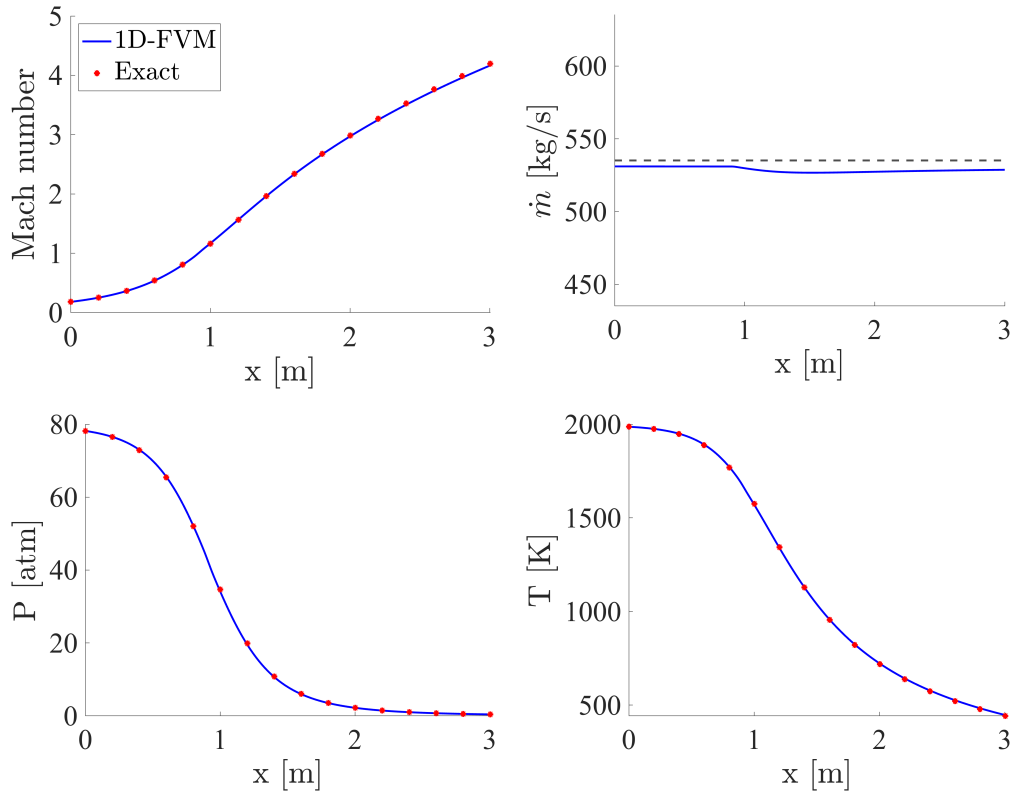


Figure C.2: Converged solution for an isentropic converging-diverging nozzle.

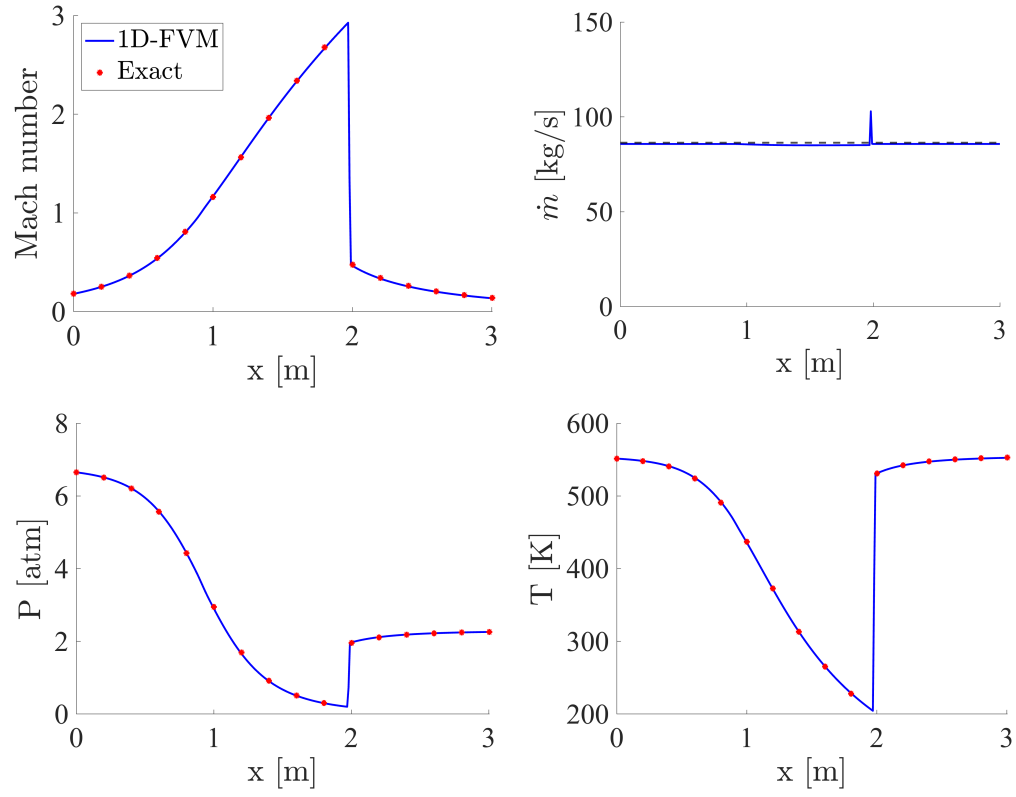


Figure C.3: Converged solution for a converging-diverging nozzle with a shock in the supersonic region.

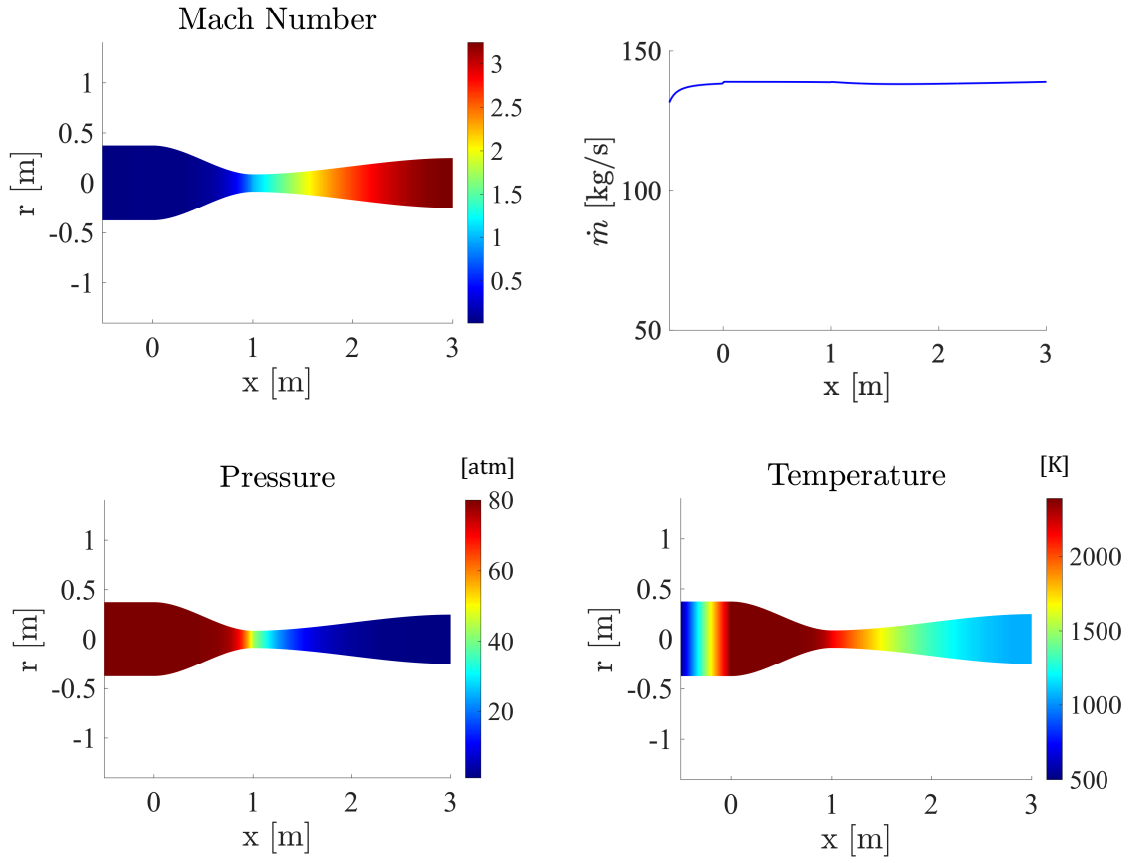


Figure C.4: Simulation of a rocket (combustor + nozzle). Chemical reactions occur from the start of the combustor ($x = -0.5$) and end at the entrance to the nozzle ($x = 0$).